

Livo Pronunciation Assessment – System Architecture

By: Purva Chopdekar

LinkedIn: www.linkedin.com/in/purva-chopdekar-631905308

Overview

Livo is a web-based pronunciation assessment application built using Next.js and Azure AI Speech Services. The application allows users to record speech, submit it for analysis, and receive real-time pronunciation feedback. The system follows a lightweight serverless architecture and is deployed on Vercel.

System Components

Frontend

The frontend is built using Next.js App Router. It handles audio recording, converts audio into WAV format using a custom encoder, sends recordings to the backend, and displays pronunciation feedback to the user.

Backend

The backend consists of a serverless API route (`/api/assess`). It receives audio from the frontend, validates requests, communicates with Azure Speech Services, and returns structured assessment results.

Azure Speech Services

Azure AI Speech Pronunciation Assessment is used as the speech evaluation engine. It provides pronunciation, accuracy, fluency, and completeness scores, along with word-level feedback on mispronounced words.

Models and APIs Used

Azure AI Speech Pronunciation Assessment API

The primary requirement of this project was pronunciation evaluation rather than simple speech-to-text conversion. Azure Speech Services provides pronunciation assessment as a built-in capability, including fluency, accuracy, completeness, and overall pronunciation scores.

Alternative transcription-focused solutions such as Whisper would have required additional logic to infer pronunciation quality from transcripts. Azure's dedicated pronunciation assessment API provided reliable scoring and reduced implementation complexity, making it the most practical choice for this project.

Pronunciation Scoring Logic

The application relies on Azure's assessment engine to generate four core metrics:

- **Accuracy:** Measures phonetic correctness.
- **Fluency:** Measures speaking rhythm and flow.
- **Completeness:** Measures whether all expected words were spoken.
- **Pronunciation Score:** Overall assessment score.

To make feedback actionable, the application highlights lower-scoring words using simple thresholds:

- Score ≥ 80 : Acceptable pronunciation
- Score 60–79: Needs improvement
- Score < 60 : Highlighted as mispronounced

This approach prioritizes the most significant pronunciation issues rather than overwhelming users with minor corrections.

DPDP Act 2023 Compliance

The application follows privacy-by-design principles aligned with India's Digital Personal Data Protection (DPDP) Act, 2023.

Consent

Users explicitly choose when to record and submit audio. No audio is collected without user action.

Storage and Retention

The current implementation uses transient processing. Audio is recorded in the browser, sent for assessment, and is not permanently stored by the application. Assessment results are displayed to the user and can be discarded after the session.

Data Residency

Speech data is processed through Azure Speech Services in the configured Azure region. For production use, an India-based or approved regional deployment should be selected where available.

Deletion

Because recordings are not persistently stored, user audio is effectively deleted after processing. If future versions introduce user accounts or saved history, user-accessible deletion mechanisms would be implemented.

Trade-offs and Future Improvements

A managed speech service was chosen to accelerate development and provide reliable pronunciation scoring. The trade-off is reduced control over the underlying assessment model and dependency on a third-party service.

The current architecture is intentionally stateless, which improves privacy and simplifies compliance but does not support long-term progress tracking.

Given additional development time, the next features would include:

- User authentication and progress history
- Pronunciation trend analytics
- AI-generated coaching recommendations
- Phoneme-level feedback
- Multi-language support
- Downloadable assessment reports

The current architecture provides a scalable, privacy-conscious foundation while maintaining a simple user experience and rapid deployment workflow.

Architecture Diagram

